Shibam **GHOSH**

RESEARCH ENGINEER, INRIA, PARIS, FRANCE

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I am a **Research Engineer** in the COSMIQ team at the French National Institute for Research in Digital Science and Technology (INRIA, Paris), working with Léo Perrin. I have completed my **PhD** thesis in the **Department of Computer Science** at the **University of Haifa**, under the supervision of Prof. Orr Dunkelman. I began my doctoral studies in October 2020. Before this, I completed my master's thesis as a research intern at **INRIA, Paris**, under the guidance of Anne Canteaut and Léo Perrin.

Research Interests

- Cryptanalysis of Symmetric-key Primitives
- Symmetric-key Primitive Design
- Lightweight Cryptography
- White-box Cryptography
- Provable Security

Education

Indian Statistical Institute — *Kolkata, India*
MASTER OF TECHNOLOGY IN CRYPTOLOGY AND SECURITY — *2018 – 2020*

Presidency University — *Kolkata, India*
MASTER OF SCIENCE IN MATHEMATICS — *2015 – 2017*

Krishnagar Government College — *Krishnagar, India*
BACHELOR OF SCIENCE IN MATHEMATICS — *2012 – 2015*

Kabi Bijoylal H. S. Institute, West Bengal Council of Higher Secondary Education — *Krishnagar, India*
HIGHER SECONDARY EDUCATION — *2010 – 2012*

Research Experience

French National Institute for Research in Digital Science and Technology (INRIA) — *Paris, France*
RESEARCH ENGINEER — *Jan 2025 – Currently*

- Supervisor: Léo Perrin
- Focus Area: Design and Analysis of Symmetric-key Primitives for advanced cryptographic protocols.

Ethereum Foundation — *Remote*
RESEARCH INTERN — *Oct 2024 – Dec 2024*

- Supervisor: Dmitry Khovratovich
- Focus Area: Analysis of post-quantum digital signature schemes based of symmetric-key primitives

University of Haifa — *Haifa, Israel*
PHD STUDENT — *Oct 2020 – Dec 2024*

- Supervisor: Prof. Orr Dunkelman
- My PhD research has centered on cryptanalysis, focusing on the design and analysis of symmetric primitives.
- A significant portion of my work is dedicated to the algebraic cryptanalysis of these primitives. Some key contributions:
  - An enhanced Fast Fourier Transform-based key-recovery attack on 6-round AES.
  - Algebraic cryptanalysis of NIST LwC candidates Ascon, KNOT, and TinyJAMBU.
  - Distinguishers derived from the structural symmetry of SHA3, Xoodyak, and the Belarusian standard bash.
  - Division property-based fault attacks on the GIFT and Present ciphers.
- In addition to cryptanalysis, I have co-designed the tweakable block cipher QARMAv2, tailored for memory protection.
- My research also extends into the area of re-keying techniques, where I have analyzed the security of the IETF/ISO standard Advanced CryptoPro Key Meshing (ACPKM) internal rekeying technique, widely used in Russian variants of TLS and CMS.

Institut national de recherche en sciences et technologies du numérique (INRIA) *Paris, France*

RESEARCH INTERN *Jan – July, 2020*

- Supervisor: Anne Canteaut, Léo Perrin
- Master's Dissertation: On the QIC of quadratic APN functions (available online here)
- My master's research focused on cryptographic Boolean functions, with particular emphasis on the Big APN Problem.
- Our goal was to generate quadratic APN functions of size ≥ 8.
- The search for APN permutations and their classification has been an open challenge for over 25 years.
- Our core approach involved representing a quadratic vectorial Boolean function using a cubic structure called a Quadratic Indicator Cube (QIC) and identifying the criteria associated with this cube that are necessary and sufficient for a function to be APN.

Publications

JOURNAL ARTICLES

1. Anup Kumar Kundu, Shibam Ghosh, Aikata Aikata, and Dhiman Saha. ToFA: Towards fault analysis of GIFT and GIFT-like ciphers leveraging truncated impossible differentials. Cryptology ePrint Archive, Paper 2024/1927, 2024 (To be appear in IACR CHES 2025)

2. Roberto Avanzi, Orr Dunkelman, and Shibam Ghosh. Differential cryptanalysis of the reduced pointer authentication code function used in arm's feat_pacqarma3 feature. *IACR Trans. Symmetric Cryptol.*, 2025(1):380–419, 2025 (DOI)

3. Marcel Nageler, Shibam Ghosh, Marlene Jüttler, and Maria Eichlseder. Autodiver: Automatically verifying differential characteristics and learning key conditions. *IACR Trans. Symmetric Cryptol.*, 2025(1):471–514, 2025 (DOI)

4. Sahiba Suryawanshi, Shibam Ghosh, Dhiman Saha, and Prathamesh Ram. Simple vs. vectorial: exploiting structural symmetry to beat the zerosum distinguisher. *Des. Codes Cryptogr.*, 93(1):133–174, 2025 (DOI).

5. Roberto Avanzi, Subhadeep Banik, Orr Dunkelman, Maria Eichlseder, Shibam Ghosh, Marcel Nageler, and Francesco Regazzoni. The QARMAv2 Family of Tweakable Block Ciphers. *IACR Transactions on Symmetric Cryptology*, 2023(3):25–73, 2023 (DOI)

6. Orr Dunkelman, Shibam Ghosh, and Eran Lambooij. Practical Related-Key Forgery Attacks on Full-Round TinyJAMBU-192/256. *IACR Trans. Symmetric Cryptol.*, 2023(2):176–188, 2023 (DOI)

7. Orr Dunkelman, Shibam Ghosh, and Eran Lambooij. Attacking the IETF/ISO Standard for Internal Re-keying CTR-ACPKM. *IACR Trans. Symmetric Cryptol.*, 2023(1):41–66, 2023 (DOI)

CONFERENCE PAPERS

1. Yanis Belkheyar, Patrick Derbez, Shibam Ghosh, Gregor Leander, Silvia Mella, Léo Perrin, Shahram Rasoolzadeh, Lukas Stennes, Siwei Sun, Gilles Van Assche, and Damian Vizár. Chilow and chichi: New constructions for code encryption. In *EUROCRYPT*, volume 15601 of *Lecture Notes in Computer Science*, pages 212–243. Springer, 2025 (DOI)

2. Orr Dunkelman, Shibam Ghosh, Nathan Keller, Gaëtan Leurent, Avichai Marmor, and Victor Mollimard. Partial Sums Meet FFT: Improved Attack on 6-Round AES. In *Advances in Cryptology – EUROCRYPT 2024*, pages 128–157. Springer, 2024 (DOI)

3. Ravi Anand, Shibam Ghosh, Takanori Isobe, and Rentaro Shiba. Quantum Key Recovery Attacks on 4-Round Iterated Even-Mansour with Two Keys. In *Proceedings of Information Security Conference (ISC)*, volume 15257 of *Lecture Notes in Computer Science*, pages 87–103. Springer, 2024 (DOI)

4. Shibam Ghosh, Anup Kumar Kundu, Mostafizar Rahman, and Dhiman Saha. Bizness: Bit invariant zero-sum property based on division trail. In *INDOCRYPT*, volume 15496 of *Lecture Notes in Computer Science*, pages 134–155. Springer, 2024 (DOI)

5. Anup Kumar Kundu, Shibam Ghosh, Dhiman Saha, and Mostafizar Rahman. Divide and Rule: DiFA - Division Property Based Fault Attacks on PRESENT and GIFT. In *ACNS (1)*, volume 13905 of *Lecture Notes in Computer Science*, pages 89–116. Springer, 2023 (DOI)

6. Orr Dunkelman, Shibam Ghosh, and Eran Lambooij. Full Round Zero-Sum Distinguishers on TinyJAMBU-128 and TinyJAMBU-192 Keyed-Permutation in the Known-Key Setting. In *INDOCRYPT*, volume 13774 of *Lecture Notes in Computer Science*, pages 349–372. Springer, 2022 (DOI)

7. Nilanjan Datta, Avijit Dutta, and Shibam Ghosh. INT-RUP Security of SAEB and TinyJAMBU. In *INDOCRYPT*, volume 13774 of *Lecture Notes in Computer Science*, pages 146–170. Springer, 2022 (DOI)

8. Shibam Ghosh and Orr Dunkelman. Automatic Search for Bit-Based Division Property. In *LATINCRYPT*, volume 12912 of *Lecture Notes in Computer Science*, pages 254–274. Springer, 2021 (DOI)

9. Shibam Ghosh and Léo Perrin. Some Experimental Results on Quadratic APN Functions. In *Boolean Functions and their Applications (BFA) 2021*, 2021 (available online here)

ePrint Archive

1. Arghya Bhattacharjee, Ritam Bhaumik, Nilanjan Datta, Avijit Dutta, Shibam Ghosh, and Sougata Mandal. BBB secure arbitrary length tweak TBC from n-bit block ciphers. Cryptology ePrint Archive, Paper 2024/2049, 2024

2. Nilanjan Datta, Avijit Dutta, Shibam Ghosh, and Hrithik Nandi. Optimally secure TBC based accordion mode. Cryptology ePrint Archive, Paper 2024/2053, 2024

Research Talks Delivered

2025	**Differential Cryptanalysis of the Arm's FEAT_PACQARMA3 Feature**, FSE, 2025	*Rome, Italy*
2024	**Partial Sums Meet FFT: Improved Attack on 6-Round AES**, Eurocrypt, 2024	*Zurich, Switzerland*
2023	**Attacking the IETF/ISO Standard for Internal Re-keying CTR-ACPKM**, FSE, 2023	*Kobe, Japan*
2022	**Full Round Zero-Sum Distinguishers on TinyJAMBU-128 and TinyJAMBU-192**, Indocrypt, 2022	*Kolkata, India*
2021	**Automatic Search for Bit-based Division Property**, Latincrypt, 2021	*Vitual*
2021	**Some Experimental Results on Quadratic APN Functions**, BFA, 2021	*Vitual*

Technical Projects

Improved Attack on 6-Round AES

Practical Attacks

- I have presented a Fast Fourier Transform-based key-recovery attack on 6-round AES at Eurocrypt 2024.
- The attack is fully implemented in C and verified on Amazon AWS servers.
- The source code is available in my git and published in IACR artifact.

Forgery Attacks on NIST Lightweight Crypto Finalist TinyJAMBU

Practical Attacks

- I have implemented a practical related-key forgery attack on the TinyJAMBU-v2 with 256/192-bit keys in C.
- Part of this work was published at FSE 2023 and the source code is available in my git.

Automatic Tools For Algebraic Attacks

Tools In Cryptanalysis

- I have prepared Pyhon-based automatic tools for cryptanalysis of NIST lightweight ciphers using MILP, SAT, and CP, available in my git.

Automatic Tool for Verifying Differential Characteristics

Tools In Cryptanalysis

- This project involves developing a tool using publicly available #SAT solvers that thoroughly verifies differential characteristics.
- The tool calculates the expected probability of differential characteristics while considering the cipher's key schedule.
- This tool also estimates the size of the key-space that validates the characteristic and deduces conditions for these keys.
- The source code of the tool is available in git.

Security Analysis of Arm's Pointer Authentication Code (PAC)

PRACTICAL ATTACKS

- This project focuses on the Qarma cipher, utilized in the Arm A-profile and M-profile architectures as a Pointer Authentication Code.
- Pointer Authentication Code (PAC) is important to mitigate Return-Oriented Programming (ROP) exploits.
- We have prepared automatic tools to search differential properties of Qarma and mounted practical attacks on various PAC versions.
- The source code of the project is available in my git.

Ongoing Projects

White-Box Secure Cipher

PRIMITIVE DESIGN

- We aim to propose an Even-Mansour variant of white-box secure cipher family
- The main objective is to achieve space-hardness and longevity security in specific adversarial environments.

Teaching Experience

University of Haifa
Haifa, Israel

TEACHING ASSISTANT
2022 – 2024

- Introduction to cryptography, Spring Semester, 2022 (lecturer in charge: Prof. Orr Dunkelman)
- Introduction to cryptography, Spring Semester, 2023 (lecturer in charge: Prof. Orr Dunkelman)
- Introduction to cryptography, Spring Semester, 2024 (lecturer in charge: Prof. Orr Dunkelman)
- Israeli School on Biometrics, 2024 (lecturer in charge: Prof. Orr Dunkelman)

Other Professional Activities

REVIEWER

- Asiacrypt 2024
- Selected Areas in Cryptography (SAC) 2025, 2024
- Designs, Codes and Cryptography (DCC) 2025, 2024, 2023, 2022

Awards

2021	**Israeli Science Foundation (ISF) Fellowship**, Israeli Science Foundation (ISF)	*Haifa, Israel*
2020	**Data Science Research Center (DSRC) Fellowship**, University of Haifa	*Haifa, Israel*
2020	**Awarded departmental-topper prize**, Indian Statistical Institute	*Kolkata, India*
2019	**Rank 88 in National Eligibility Test**, Council of Scientific and Industrial Research (CSIR), India	*Kolkata, India*

Technical Skills

- C, C++ ●●●●○
- python, SageMath ●●●●○
- Rust ●●●○○
- Git ●●●●○
- Qiskit ●●●○○

Languages

Native Bengali
Fluent English, Hindi